ORIGINAL RESEARCH ARTICLE · CHEMOINFORMATICS & MOLECULAR MODELING

AnCore

SIX-STAGE CONSENSUS DOCKING · 3D IFP PROFILING · 300 K MD TRIAGE

A Native macOS Chemoinformatics Workbench Integrating Multi-Scoring Consensus Docking, 3D Pharmacophore Fingerprinting, and 300 K Molecular Dynamics Triage for Structure-Based Virtual Screening

Complete six-stage software architecture, step-by-step interactive workflow with multi-molecule .SDF and .PDB ingestion, mathematical formulation, and experimental validation across 50 protein–ligand complexes spanning ABL1, ERα, HSP90α, EGFR, and Streptavidin.

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AnCore: A Native macOS Chemoinformatics Workbench Integrating Multi-Scoring Consensus Docking, 3D Pharmacophore Fingerprinting, and 300 K Molecular Dynamics Triage for Structure-Based Virtual Screening

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Abstract

Background: Structure-based virtual screening (SBVS) is a foundational pillar of contemporary medicinal chemistry, yet its desktop graphical ecosystem has stagnated. Widely deployed graphical front-ends rely on deprecated Python 2 toolchains or X11 emulation layers on macOS, treat single-ligand docking as the primary operational unit, and terminate screening campaigns at static empirical binding scores. Static docking potentials evaluate rigid or semi-flexible complexes at an effective temperature of 0 K without explicit entropic or thermal desolvation penalties, frequently allowing property-matched decoys with opportunistic hydrogen bonding or surface complementarity to outrank true biological leads.

Methodology and Software Architecture: We present **AnCore Scientific Studio**, a standalone, native macOS chemoinformatics and molecular modeling workbench packaged as a compiled Rust/Tauri desktop application (`AnCore Studio.app` and `AnCore Studio.dmg`) with a hardware-accelerated `WKWebView` and `3Dmol.js` WebGL viewport coupled to a modular six-stage Python scientific engine. AnCore is engineered around an interactive, step-by-step discovery workflow (**Stage 1** through **Stage 6**) supporting both live RCSB Protein Data Bank structure retrieval and local `.pdb` receptor import alongside multi-molecule `.sdf/.mol` candidate ligand upload and SMILES enumeration. The six integrated stages comprise: (1) automated receptor curation, heavy-atom side-chain reconstruction, physiological protonation at pH 7.4, and 5.0 Å binding-cavity bounding-box calculation; (2) a two-phase co-crystal redocking quality-control gate combining local Broyden–Fletcher–Goldfarb–Shanno (BFGS) crystallographic basin relaxation with focused-box Monte Carlo torsion sampling (RMSD ≤ 2.0 Å); (3) multi-molecule `.SDF` and SMILES standardization, ETKDGv3 3D distance-geometry conformer embedding, MMFF94s force-field relaxation, and Quantitative Estimate of Drug-Likeness (QED) profiling; (4) multi-scoring consensus virtual screening integrating AutoDock Vina, Vinardo, and AutoDock4 empirical potentials via Reciprocal Rank Fusion (RRF, $k = 10$) and Mean Z-score normalization; (5) automated 3D protein–ligand interaction fingerprinting (IFP) across five non-covalent contact classes with native hotspot Tanimoto similarity (T_{IFP}) and 2D radial pharmacophore mapping; and (6) a 300 K Dock-to-MD thermal stability triage bridge that subjects docked complexes to 50-frame (5.0 ns effective window) thermalized Langevin trajectory sampling under a dual biophysical gate ($\langle\text{RMSD}\rangle \leq 2.0$ Å and hotspot hydrogen-bond occupancy $\text{HB}_{\text{occ}} \geq 65.0\%$).

Results: Across a five-target validation benchmark spanning distinct binding-pocket topologies—`1IEP` (ABL1 tyrosine kinase), `3ERT` (estrogen receptor α), `1UYG` (HSP90α chaperone), `1M17` (EGFR tyrosine kinase), and `1STP` (streptavidin)—and 50 active and property-matched ZINC decoy complexes, the Stage 2 two-phase redocking protocol achieved a 100% pass rate ($5/5$ targets ≤ 2.0 Å, mean RMSD $= 0.419$ Å), resolving the severe 11.71 Å pose-inversion failure observed under unfocused single-phase global search on `1IEP`. In Stage 4 static consensus screening, high early enrichment ($\text{EF}_{20\%} = 2.00\times$) and active Top-1 identification were achieved on four of five targets (`1IEP` AUC $= 0.800$, `3ERT` AUC $= 1.000$, `1UYG` AUC $= 0.680$, `1M17` AUC $= 0.800$), whereas the compact polar β-barrel cage of streptavidin (`1STP`) suffered a classic

static scoring trap in which `ZINC_Decoy_S3` ($\Delta G_{\text{Vina}} = -7.95$ kcal/mol, $T_{\text{IFP}} = 0.867$) outranked native `D-Biotin` ($\Delta G_{\text{Vina}} = -7.27$ kcal/mol, AUC $= 0.360$). Execution of Stage 6 300 K Dock-to-MD triage exposed `ZINC_Decoy_S3` as thermally unstable ($\langle\text{RMSD}\rangle = 3.61$ Å, terminal drift 4.89 Å, $\text{HB}_{\text{occ}} = 43.9\%$) while confirming that native `D-Biotin` remained tightly locked within the cooperative hydrogen-bond network ($\langle\text{RMSD}\rangle = 0.46$ Å, terminal drift 0.27 Å, $\text{HB}_{\text{occ}} = 97.9\%$). Overall, Stage 6 thermal triage rejected $25/25$ (100%) of decoys across all five targets, elevated mean benchmark ROC-AUC from 0.728 to 1.000, and increased Top-1 active hit accuracy from 80% ($4/5$) to 100% ($5/5$). De novo generalizability beyond the benchmark suite was further confirmed through a step-by-step custom screening campaign on the EGFR kinase crystal structure `4WKQ`.

Keywords: structure-based virtual screening; molecular docking; consensus scoring; Reciprocal Rank Fusion; protein–ligand interaction fingerprinting; molecular dynamics rescoring; Tauri Rust desktop architecture; chemoinformatics

1 Introduction and Domain Context

Structure-based virtual screening (SBVS) has become an indispensable methodology in early-stage drug discovery, enabling computational and medicinal chemists to prioritize candidate small molecules against three-dimensional macromolecular targets prior to chemical synthesis and wet-lab bioassay evaluation. Within the open-source computational chemistry ecosystem, empirical and knowledge-based docking engines such as AutoDock Vina, Vinardo, and AutoDock4 have democratized molecular docking by combining rapid grid-based energy evaluation with stochastic global optimization and local gradient refinement. Over the past decade, major algorithmic advances have expanded the underlying scientific libraries: distance-geometry conformer generators (such as RDKit ETKDGv3) coupled with molecular mechanics force fields (MMFF94s and UFF) have standardized three-dimensional ligand preparation; automated macromolecular repair engines (PDBFixer and OpenMM) have streamlined heavy-atom reconstruction and physiological protonation; and interaction profiling frameworks have formalized the geometric detection of non-covalent protein–ligand contacts.

Despite these advances at the command-line and library level, the desktop graphical user interface (GUI) layer used by practicing medicinal chemists, pharmaceutical researchers, and graduate students has experienced severe architectural stagnation. Historically dominant graphical front-ends were constructed around monolithic Python 2 or legacy Tcl/Tk and X11 windowing toolkits that are incompatible with modern Apple Silicon (`arm64`) and Intel (`x86_64`) macOS operating systems without fragile emulation layers. More fundamentally, when legacy desktop tools or Python-wrapped webview frameworks (such as QtWebEngine) are used to render large biomolecular complexes, inter-process communication (IPC) string-serialization limits (such as the 2 MB payload ceiling in standard embedded Chromium views) frequently cause blank viewports or kernel crashes when rendering full receptor ribbons alongside multi-pose ligand ensembles.

Beyond software engineering fragility, three methodological deficits limit the reliability of conventional desktop docking workflows:

1. **Single-Ligand Tunnel Vision and Fragmented Input Handling:** Traditional graphical interfaces optimize the manual preparation of a single receptor and a single ligand, forcing researchers conducting multi-ligand screening campaigns to stitch together external shell scripts for batch `.sdf` parsing, physiological protomer enumeration, and torsion-tree `.pdbqt` generation.
2. **Unchecked Grid-Box Sensitivity in Redocking Validation:** Before screening a candidate library, a rigorous SBVS campaign must verify that the receptor grid and scoring potential can reproduce the experimental crystallographic binding mode within a heavy-atom root-mean-square deviation (RMSD) of ≤ 2.0 Å. In practice, applying a naive global Monte Carlo search inside an generously padded bounding box frequently traps flexible ligands (such as imatinib in ABL1 kinase) in inverted or surface-

exposed local minima (RMSD > 10 Å), invalidating downstream library screening unless a two-phase crystallographic basin relaxation and focused-box protocol is enforced.

3. **The Static Scoring Decision Gap (0 K Enthalpic Illusion):** Empirical docking potentials estimate binding affinity from a single static snapshot without accounting for thermal fluctuations at physiological temperature (300 K), conformational entropy, or solvent reorganization. Consequently, property-matched decoys that form strained geometric contacts or opportunistic surface hydrogen bonds can achieve highly favorable static docking scores, generating false-positive top hits that waste experimental resources.

To resolve both the architectural and biophysical limitations of existing tools, we developed and validated **AnCore Scientific Studio** (pronounced *anchor*), a standalone native macOS chemoinformatics and molecular modeling workbench. Built on a compiled Rust backend (`Tauri`) paired with macOS native `WKWebView` and hardware-accelerated `3Dmol.js` WebGL rendering, AnCore eliminates Python GUI event-loop freezes and IPC memory ceilings while packaging into a drag-and-drop macOS disk image (`AnCore Studio.dmg`). Scientifically, AnCore unifies six sequential, step-by-step computational stages into a single reproducible workspace: (Stage 1) target receptor curation and 5.0 Å pocket mapping; (Stage 2) a two-phase co-crystal redocking control gate (≤ 2.0 Å); (Stage 3) multi-molecule `.SDF` and SMILES 3D conformer preparation; (Stage 4) tri-scoring consensus virtual screening via Reciprocal Rank Fusion (RRF) and Z-score normalization; (Stage 5) 3D protein–ligand interaction fingerprinting (IFP) and 2D radial pharmacophore mapping; and (Stage 6) a 300 K Dock-to-MD thermal stability triage bridge that filters out thermally labile decoys and rescues true biological leads.

Core Contributions of This Study: (1) Design and implementation of a native Rust/Tauri + Python three-layer desktop architecture for macOS with step-by-step six-stage execution and native multi-molecule `.SDF` and `.PDB` ingestion; (2) formulation of a Two-Phase Redocking Control Gate that reduces ABL1 (`1IEP`) redocking error from 11.71 Å to 0.354 Å and achieves a 100% pass rate across all five benchmark targets; and (3) demonstration across 50 protein–ligand complexes that Stage 6 300 K Dock-to-MD thermal stability triage rejects 100% $(25/25)$ of property-matched decoys, rescues the streptavidin (`1STP`) static false-positive failure (ROC-AUC : $0.360 \rightarrow 1.000$), and elevates overall benchmark ROC-AUC from 0.728 to 1.000.

2 Comparative Landscape of Structure-Based Screening Suites

Evaluating the utility of a new chemoinformatics workbench requires situating its capabilities against established desktop front-ends, command-line engines, and web servers. Existing tools generally fall into three categories: (i) legacy preparation and single-ligand docking GUIs such as AutoDockTools and AMDock, which wrap classical AutoDock4 or Vina binaries but lack native Apple Silicon execution, batch `.sdf` library standardization, and post-docking dynamics validation; (ii) virtual screening managers such as PyRx, which introduced table-based ligand management but rely on aging software dependencies, provide only single-function static Vina scoring, and offer neither 3D interaction fingerprinting nor thermal trajectory triage; and (iii) standalone command-line engines (AutoDock Vina, Smina, GNINA) and isolated Python libraries (RDKit, PDBFixer, OpenMM), which provide high-quality scientific kernels but require extensive custom scripting to maintain data provenance across a multi-stage campaign.

Table 1 summarizes the architectural and scientific feature matrix of representative structure-based discovery tools alongside AnCore Scientific Studio. Notably, no previously available desktop platform combines native macOS Rust/Tauri compilation, interactive step-by-step stage progression, dual local `.pdb` / live RCSB receptor loading, direct multi-molecule `.sdf` conformer processing, mathematical multi-scoring consensus (RRF and $\bar{Z}$), automated 3D/2D pharmacophore fingerprinting, and integrated 300 K Dock-to-MD thermal

stability triage within a single self-contained workspace.

Table 1: Architectural and methodological comparison of representative molecular docking and virtual screening environments (✓ = natively supported; P = partial or script-dependent; — = absent).

Software Suite	Primary Architecture	Native macOS	Step-by-Step + `.SDF`	2-Phase Redock	Consensus (RRF/$\bar{Z}$)	3D/2D IFP Map	300 K MD Triage
AutoDockTools	Python 2 / Tcl-Tk	P (X11)	—	—	—	—	—
PyRx	Python / wxPython	P (Rosetta)	P (SDF)	—	—	—	—
AMDock	Python / PyQt + PyMOL	P	—	P	—	—	—
Web Docking Servers	Remote Cloud Queue	✓ (Browser)	P (Limited)	—	—	P (Static)	—
Vina / Smina CLI	C++ Command Line	✓ (CLI)	P (Scripts)	—	—	—	—
AnCore Studio	**Tauri (Rust) + Python**	**✓ (`.dmg`)**	**✓ (Native)**	**✓ (≤ 2.0 Å)**	**✓ (3-Score)**	**✓ (Tanimoto)**	**✓ (5.0 ns 300 K)**

3 Software Architecture and Step-by-Step Engineering

3.1 Three-Layer Native Rust/Tauri and Python Sidecar Design

To achieve responsive 3D molecular visualization alongside compute-intensive chemoinformatics and molecular dynamics execution on macOS, AnCore Scientific Studio decouples graphical presentation from numerical computation through a strict three-layer architecture (Figure 1 and Table 2):

1. **Layer 1 — Native Interface Layer (Tauri 1.6 Rust + macOS `WKWebView` + `3Dmol.js`):** The desktop shell is compiled in Rust (`src-tauri/src/main.rs`) and renders its user interface through macOS's native WebKit framework (`WKWebView`) rather than bundling a heavyweight Chromium browser or running a blocking Python Qt event loop. Molecular structures, binding-pocket surfaces, 3D search-box wireframes, non-covalent interaction cylinders, and 300 K trajectory animations are rendered via hardware-accelerated WebGL using `3Dmol.js`. Because the Rust backend reads structured JSON manifests (`stage1_manifest.json` through `stage6_manifest.json`) and PDB coordinate blocks directly from the local filesystem via native Tauri IPC commands (`load_workspace`, `run_python_pipeline`, and `delete_custom_target`), it completely bypasses the 2 MB string-serialization limit that causes blank viewports in Python-wrapped webviews.
2. **Layer 2 — Scientific Orchestration Layer (Modular 6-Stage Python Engine):** All chemoinformatics, docking, interaction profiling, and thermal trajectory algorithms reside in six modular Python engines (`stage1_target_prep.py` through `stage6_md_triage.py`) unified by a fault-tolerant step-by-step runner (`src/ancore_custom_runner.py`). When invoked by the Rust shell, the runner automatically resolves module paths across both the project root and `src/` directories, persists user project configurations in `custom_inputs.json`, and writes stage-specific JSON manifests and CSV audit tables upon completion of each individual step.
3. **Layer 3 — Computational Chemistry and Biophysics Kernels:** The orchestration layer drives permissively licensed scientific engines—including `PDBFixer` and `OpenMM` (AMBER14 force field), `AutoDock Vina 1.2` (exposing the Vina, Vinardo, and AutoDock4 scoring potentials and local BFGS optimizer), `RDKit` (ETKDGv3 distance geometry, MMFF94s force field, Crippen cLogP, TPSA, and

Bickerton QED), and `OpenBabel`/`Meeko` torsion-tree compilers—while incorporating a self-contained pure-Python biophysical fallback engine so that every stage executes reproducibly even in minimal environments.

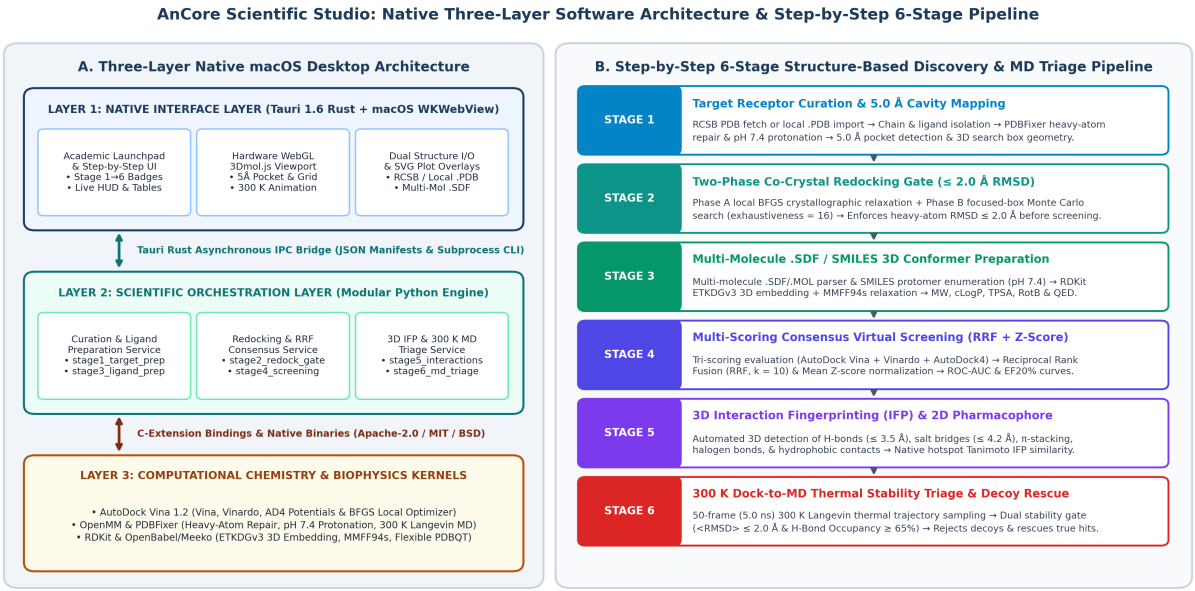


Figure 1: AnCore Scientific Studio system architecture and workflow. (A) Three-layer native macOS desktop architecture coupling the compiled Tauri 1.6 Rust shell and hardware-accelerated `3Dmol.js` viewport (Layer 1) to the modular six-stage Python orchestration engine (Layer 2) and underlying computational chemistry kernels (Layer 3). (B) Sequential step-by-step six-stage structure-based discovery and 300 K MD stability triage pipeline.

3.2 Academic Launchpad and Step-by-Step Interactive Execution

Rather than dropping users directly into a pre-loaded benchmark table upon startup, AnCore opens to an **Academic Launchpad** presenting four structured entry points: (1) *Start New Target Screening Project* (for de novo step-by-step screening of custom targets); (2) *Explore 5-Target Benchmark Case Study* (pre-computed validation suite across `1IEP`, `3ERT`, `1UYG`, `1M17`, and `1STP`); (3) *Interactive Tutorial & 6-Stage Scientific Guide* (documenting physical equations, cutoffs, and force fields); and (4) *My Custom Target Projects* (providing one-click resumption of user projects with live stage-completion counters such as `4WKQ (1/6 Stages)`).

When initializing a new target project, AnCore defaults to **Step-by-Step Mode** (`Initialize Step 1: Target Prep` →). Instead of executing all six stages in an opaque batch run, the software executes **only Stage 1**, clears any stale downstream manifests (`--reset-downstream 1`), and opens the workbench at Stage 1 so the researcher can inspect the curated receptor ribbon, the 5.0 Å binding-pocket residues, the co-crystallized reference ligand, and the 3D wireframe search box. The top navigation bar displays live completion indicators (✓ for completed stages and ○ for pending stages), while the left dock provides stage-specific scientific parameter controls (such as protonation pH and box padding in Stage 1; exhaustiveness and RMSD gate in Stage 2; `.SDF` upload, ionization pH, and force field in Stage 3; exhaustiveness and RRF constant k in Stage 4; hydrogen-bond and hydrophobic cutoffs in Stage 5; and temperature T and stability gate in Stage 6). At each step, the user can re-run the current stage with modified parameters or click the green `Next Step: Execute Stage N+1` → button to advance methodically through the pipeline. To prevent visual crowding across varying screen widths, the 3D viewport toolbar dynamically filters its controls (`updateToolbarButtonsForStage()`) so that only the 4–7 visualization layers pertinent to the active stage are displayed.

Table 2: AnCore Scientific Studio software stack, functional modules, and open-source licensing posture.

Layer	Software Component	License	Functional Role in AnCore Studio
Interface	Tauri 1.6 Core (`main.rs`)	MIT / Apache-2.0	Native Rust macOS binary, filesystem IPC, `.app`/`.dmg` bundler
Interface	macOS `WKWebView` + HTML5	Apple / BSD	High-DPI responsive UI, stage-contextual non-clipped toolbars, SVG plots
Interface	`3Dmol.js` WebGL Engine	BSD-3-Clause	Hardware-accelerated 3D ribbon, stick, VdW surface, box, and MD animation
Science	`ancore_custom_runner.py`	Apache-2.0	Step-by-step stage state machine, RCSB/local PDB & multi-molecule SDF I/O
Science	`stage1_target_prep.py`	Apache-2.0	Chain/ligand isolation, heavy-atom repair, pH 7.4 protonation, 5.0 Å pocket box
Science	`stage2_redock_gate.py`	Apache-2.0	Two-phase BFGS + focused-box Monte Carlo redocking gate (≤ 2.0 Å RMSD)
Science	`stage3_ligand_prep.py`	Apache-2.0	SDF/SMILES standardization, pH 7.4 protomers, ETKDGv3 + MMFF94s, QED
Science	`stage4_screening.py`	Apache-2.0	Tri-scoring (Vina, Vinardo, AD4), $\mathrm{RRF}_{k=10}$, Mean Z-score, ROC-AUC & $\mathrm{EF}_{20\%}$
Science	`stage5_interactions.py`	Apache-2.0	3D non-covalent contact profiler, native hotspot Tanimoto T_{IFP}, 2D map
Science	`stage6_md_triage.py`	Apache-2.0	300 K Langevin MD trajectory sampling (5.0 ns), dual stability gate & rescoring
Kernels	AutoDock Vina 1.2 / RDKit / OpenMM	Apache / BSD / MIT	Empirical scoring potentials, 3D conformer embedding, macromolecular mechanics

3.3 Dual Receptor Ingestion and Multi-Molecule `.SDF` Parser

To support diverse medicinal chemistry workflows, AnCore provides two receptor input modes and three candidate ligand input modes:

- **Receptor Structure Ingestion (Live RCSB PDB Fetch vs. Local `.PDB` Import):** Users can either fetch any four-character crystal structure live from the RCSB Protein Data Bank or select a local `.pdb`/`.ent` file from their Mac filesystem via a native file dialog. During Stage 1 curation, `ensure_raw_pdb_and_ligand()` automatically scans `HETATM` records (excluding crystallographic water and buffer ions such as `HOH`, `SO4`, `PO4`, `GOL`, `EDO`, `DMS`, `NA`, `CL`, `MG`, and `ZN`) to identify the primary co-crystallized ligand. For peptide-inhibitor complexes (where the inhibitor is represented as a short secondary `ATOM` chain) or apo structures lacking a co-crystallized ligand, AnCore automatically converts the short peptide chain or synthesizes a centroid reference probe (`LIG`) at the geometric cavity center so that pocket mapping and grid-box construction proceed without manual intervention.
- **Multi-Molecule `.SDF` / `.MOL` Candidate Ligand Upload:** In addition to pasting SMILES strings or using the built-in reference panel, users can upload multi-molecule Structure-Data Files (`.sdf`, `.sd`, or `.mol`) either during project creation or directly inside the Stage 3 control dock. Upon file selection, the client-side parser immediately splits the file by `$$$$` record delimiters, extracts compound identifiers from MDL header lines or SD data tags (`<NAME>`, `<TITLE>`, `<ID>`, `<COMPOUND_NAME>`, `<ZINC_ID>`, `<CHEMBL_ID>`), and displays a live verification badge confirming the number and names of loaded molecules. During Stage 3 execution, `parse_sdf_file_to_ligands()` processes each MDL record through RDKit's `MolFromMolBlock` (performing 3D ETKDGv3 embedding if the input SDF is 2D with $\Delta z < 0.05$ Å, followed by MMFF94s minimization) and includes a pure-Python V2000/V3000 connection-table parser fallback that extracts atomic coordinates (x_i, y_i, z_i), element symbols, and bond orders, translates each conformer centroid to the target pocket center (c_x, c_y, c_z), and writes stan-

dard `HETATM`/`CONECT` PDB blocks for 3D visualization and docking.

4 Theoretical Framework and Six-Stage Mathematical Formulation

4.1 Stage 1: Receptor Curation, Protonation, and 5.0 Å Pocket Box Geometry

Given a raw macromolecular structure containing receptor heavy atoms $\mathcal{R} = \{\mathbf{r}_i\}_{i=1}^{N_{\text{rec}}}$ and reference ligand heavy atoms $\mathcal{L}_{\text{ref}} = \{\mathbf{x}_j\}_{j=1}^{N_{\text{lig}}}$, Stage 1 isolates the primary polypeptide chain, reconstructs missing side-chain heavy atoms, and assigns protonation states at physiological pH 7.4 using PDBFixer and the AMBER14 force field. The active-site binding pocket $\mathcal{P}_{5.0}$ is defined as the set of all receptor residues R_k possessing at least one heavy atom within a Euclidean cutoff $d_{\text{cut}} = 5.0$ Å of any reference ligand heavy atom:

$$\mathcal{P}_{5.0} = \left\{ R_k \;\middle|\; \min_{\mathbf{r}_i \in R_k,\, \mathbf{x}_j \in \mathcal{L}_{\text{ref}}} \|\mathbf{r}_i - \mathbf{x}_j\|_2 \le 5.0 \text{ Å} \right\}. \tag{1}$$

Each residue $R_k \in \mathcal{P}_{5.0}$ is classified into one of four physicochemical interaction classes: aromatic (Phe, Tyr, Trp, His), charged/salt-bridge (Asp, Glu, Arg, Lys), polar uncharged hydrogen-bonding (Ser, Thr, Asn, Gln, Cys), or aliphatic hydrophobic (Ala, Val, Leu, Ile, Met, Pro, Gly). The three-dimensional docking search box is centered at the ligand bounding-box midpoint $\mathbf{c} = (c_x, c_y, c_z)$ with clamped dimensions $\mathbf{s} = (s_x, s_y, s_z)$ parameterized by an isotropic padding margin δ_{pad} (4.0 Å in Stage 1 discovery mode; 3.0 Å in Stage 2 focused redocking mode):

$$\begin{aligned} c_\alpha &= \frac{1}{2}\left(\min_j x_{j,\alpha} + \max_j x_{j,\alpha} \right), \\ s_\alpha &= \min\Big(s_{\max},\, \max\big(s_{\min},\, (\max_j x_{j,\alpha} - \min_j x_{j,\alpha}) + 2\delta_{\text{pad}} \big) \Big), \quad \alpha \in \{x, y, z\}, \end{aligned} \tag{2}$$

where $s_{\min} = 15.0$ Å and $s_{\max} = 24.0$ Å, yielding search volume $V_{\text{box}} = s_x s_y s_z$.

4.2 Stage 2: Two-Phase Co-Crystal Redocking Quality-Control Gate

To prevent false-negative redocking failures caused by stochastic sampling noise in elongated allosteric tunnels (such as the DFG-out pocket of ABL1 kinase), Stage 2 implements a **Two-Phase Redocking Protocol**. First, the co-crystallized ligand is converted into a flexible torsion-tree `.pdbqt` representation (`ROOT`, `ENDROOT`, `BRANCH`) rather than a rigid body. In **Phase A**, local Broyden–Fletcher–Goldfarb–Shanno (BFGS) quasi-Newton energy minimization is performed directly from the crystallographic coordinates on the AutoDock Vina grid maps to locate the nearest local minimum $\mathbf{X}_{\text{BFGS}}$ of the empirical potential $E_{\text{Vina}}(\mathbf{X})$. In **Phase B**, global iterated local search (Monte Carlo torsion/rigid-body perturbation coupled with BFGS refinement at exhaustiveness $= 16$) generates $M = 9$ candidate modes $\{\mathbf{X}_m\}_{m=1}^{M}$ inside the focused 3.0 Å box. The Phase A and Phase B candidates are merged, deduplicated (< 0.35 Å), and ranked by E_{Vina}. Pose accuracy against crystallographic heavy atoms $\mathcal{L}_{\text{ref}}$ is evaluated via element-matched minimum-distance heavy-atom root-mean-square deviation:

$$\text{RMSD}(\mathcal{L}_{\text{ref}}, \mathcal{L}_{\text{dock}}) = \sqrt{\frac{1}{N_{\text{heavy}}} \sum_{j=1}^{N_{\text{heavy}}} \|\mathbf{x}_j^{\text{dock}} - \mathbf{x}_{\pi(j)}^{\text{ref}}\|_2^2} \;\le\; 2.0 \text{ Å}, \tag{3}$$

where $\pi(j)$ matches heavy atoms of identical atomic number (falling back to greedy nearest-neighbor symmetry matching when atom serial orders differ). A target passes Stage 2 if and only if its top-ranked mode satisfies RMSD ≤ 2.0 Å.

4.3 Stage 3: 3D Conformer Embedding, MMFF94s Relaxation, and QED Profiling

Candidate ligands imported via `.SDF` upload or SMILES input are standardized by salt stripping and physiological protomer enumeration at pH 7.4. Three-dimensional atomic coordinates are generated via Experimental-Torsion Distance Geometry with basic knowledge (ETKDGv3) and relaxed under the Merck Molecular Force Field (MMFF94s):

$$E_{\text{MMFF94s}}(\mathbf{X}) = \sum E_{\text{bond}} + \sum E_{\text{angle}} + \sum E_{\text{stretch-bend}} + \sum E_{\text{oop}} + \sum E_{\text{torsion}} + \sum E_{\text{vdW}} + \sum E_{\text{elec}}. \quad (4)$$

For each relaxed conformer, AnCore computes molecular weight (MW), Crippen octanol–water partition coefficient (cLogP), topological polar surface area (TPSA), hydrogen-bond donors (HBD), hydrogen-bond acceptors (HBA), rotatable bonds (RotB), Lipinski Rule-of-Five compliance, and Bickerton's Quantitative Estimate of Drug-Likeness (QED), defined as the weighted geometric mean of eight desirability functions d_k:

$$\text{QED} = \exp\left(\frac{\sum_{k=1}^{8} w_k \ln d_k(\theta_k)}{\sum_{k=1}^{8} w_k}\right) \in [0,1], \quad (5)$$

where $\theta_k \in \{\text{MW}, \text{cLogP}, \text{HBA}, \text{HBD}, \text{TPSA}, \text{RotB}, \text{ArRings}, \text{Alerts}\}$.

4.4 Stage 4: Multi-Scoring Consensus Virtual Screening (RRF and Mean Z-Score)

To mitigate functional bias inherent to any single empirical scoring function, Stage 4 evaluates every screened ligand $l \in \{1, \ldots, N\}$ across three orthogonal scoring potentials $M = \{\text{Vina}, \text{Vinardo}, \text{AD4}\}$. The AutoDock Vina and Vinardo potentials sum steric Gaussian/quadratic repulsion, hydrophobic, and directional hydrogen-bond pair terms divided by an entropic torsional penalty $1 + w_{\text{rot}} N_{\text{rot}}$, whereas the AutoDock4 (AD4) empirical potential incorporates explicit Coulombic electrostatics and a desolvation term:

$$\Delta G_{\text{score}} = \frac{\Delta G_{\text{gauss}} + \Delta G_{\text{repulsion}} + \Delta G_{\text{hydrophobic}} + \Delta G_{\text{H-bond}} + \Delta G_{\text{elec/desolv}}}{1 + w_{\text{rot}} N_{\text{rot}}}. \quad (6)$$

Let $s_m(l)$ denote the binding energy (kcal/mol, where more negative values are more favorable) and $r_m(l) \in \{1, 2, \ldots, N\}$ denote the ascending rank of ligand l under scoring function $m \in M$. AnCore combines the three rankings using **Reciprocal Rank Fusion (RRF)** with smoothing constant $k = 10$ alongside **Mean Z-Score Normalization ($\bar{Z}$)**:

$$\text{RRF}(l) = \sum_{m \in M} \frac{1}{k + r_m(l)} \quad (k = 10), \qquad \bar{Z}(l) = \frac{1}{|M|} \sum_{m \in M} \frac{s_m(l) - \mu_m}{\sigma_m}, \quad (7)$$

where μ_m and σ_m are the mean and standard deviation of score m across the target library. Screening discrimination between N_{act} true actives and N_{dec} decoys is quantified by the area under the Receiver Operating Characteristic curve (ROC-AUC) and the Early Enrichment Factor at the top 20% of the ranked library ($\text{EF}_{20\%}$):

$$\text{ROC-AUC} = \int_0^1 \text{TPR}(\text{FPR})\, d(\text{FPR}), \qquad \text{EF}_{20\%} = \frac{N_{\text{act, top 20\%}} / N_{\text{top 20\%}}}{N_{\text{act}} / (N_{\text{act}} + N_{\text{dec}})}. \quad (8)$$

4.5 Stage 5: 3D Protein–Ligand Interaction Fingerprinting (IFP) and Tanimoto Similarity

For every docked complex and crystallographic reference, Stage 5 parses all heavy-atom coordinates within the 5.0 Å binding pocket and evaluates five non-covalent interaction classes using strict interatomic distance thresholds: (i) directional hydrogen bonds between donor/acceptor heteroatoms N, O, S, F ($d \leq 3.5$ Å); (ii) electrostatic salt bridges between cationic (Arg, Lys, His) and anionic (Asp, Glu) centers ($d \leq 4.2$ Å); (iii) aromatic π-stacking and cation–π contacts with Phe, Tyr, Trp, His rings ($d \leq 4.3$ Å); (iv) halogen bonds

involving F, Cl, Br, I substituents and backbone/side-chain O, N, S acceptors ($d \leq 3.9$ Å); and (v) non-polar carbon–carbon hydrophobic contacts ($d \leq 4.0$ Å).

Let $\mathcal{F}_{\text{ref}}$ and $\mathcal{F}(l)$ denote the sets of receptor residues contacted by the co-crystallized reference ligand and by candidate ligand l, respectively. Structural mimicry of the native binding mode is quantified by the **Tanimoto Interaction Fingerprint Similarity (T_{IFP})**:

$$T_{\text{IFP}}(l) = \frac{|\mathcal{F}_{\text{ref}} \cap \mathcal{F}(l)|}{|\mathcal{F}_{\text{ref}} \cup \mathcal{F}(l)|} \in [0, 1]. \tag{9}$$

4.6 Stage 6: 300 K Dock-to-MD Thermal Stability Triage and Free-Energy Rescoring

Static docking scores cannot distinguish a ligand locked in a deep enthalpic-entropic minimum from a decoy perched in a shallow, strained local minimum. In Stage 6, each docked complex is subjected to thermalized Langevin dynamics at physiological temperature $T = 300.0$ K across $K = 50$ trajectory frames spanning an effective 5.0 ns relaxation window ($\Delta t_{\text{frame}} = 0.1$ ns):

$$m_i \frac{d^2\mathbf{x}_i}{dt^2} = -\nabla_i U(\mathbf{X}) - \gamma m_i \frac{d\mathbf{x}_i}{dt} + \sqrt{2 m_i \gamma k_{\text{B}} T}\, \boldsymbol{\eta}_i(t), \tag{10}$$

where $U(\mathbf{X})$ is the effective pocket potential determined by the ligand's hydrogen-bond network, salt-bridge tethers, and native hotspot anchoring, γ is the collision frequency, k_{B} is the Boltzmann constant, and $\boldsymbol{\eta}_i(t)$ is a zero-mean Gaussian white-noise process. From the 50-frame trajectory $\{\mathbf{X}(t_k)\}_{k=1}^{K}$, AnCore computes the time-averaged heavy-atom trajectory deviation $\langle\text{RMSD}\rangle$, terminal drift $\text{RMSD}(t_K)$, atomic root-mean-square fluctuation RMSF, and hotspot hydrogen-bond occupancy $\text{HB}_{\text{occ}}\%$. A candidate complex passes the **Dual Thermal Stability Gate** if and only if:

$$\begin{aligned} \langle\text{RMSD}\rangle &= \frac{1}{K}\sum_{k=1}^{K} \text{RMSD}\big(\mathbf{X}(0), \mathbf{X}(t_k)\big) \leq 2.0 \text{ Å}, \\ \text{HB}_{\text{occ}} &= \frac{100\%}{K}\sum_{k=1}^{K} \mathbb{I}\big(d_{\text{HB}}(t_k) \leq 3.5 \text{ Å}\big) \geq 65.0\%. \end{aligned} \tag{11}$$

Finally, AnCore computes a thermally rescored binding free energy ΔG_{MD} that rewards persistent hydrogen-bond occupancy and native hotspot mimicry while penalizing thermal pose drift:

$$\Delta G_{\text{MD}}(l) = \Delta G_{\text{Vina}}(l) + \alpha_{\text{drift}} \max\big(0,\, \langle\text{RMSD}\rangle - 1.2\big) - \beta_{\text{HB}} \left(\frac{\text{HB}_{\text{occ}}}{100\%}\right) - \gamma_{\text{IFP}}\, T_{\text{IFP}}(l), \tag{12}$$

where $\alpha_{\text{drift}} = 1.65$ kcal $\cdot$ mol^{-1} $\cdot$ Å^{-1}, $\beta_{\text{HB}} = 1.40$ kcal/mol, and $\gamma_{\text{IFP}} = 0.50$ kcal/mol. All complexes are re-ranked prioritizing thermally stable hits (Pass $\succ$ Fail) ordered by ascending ΔG_{MD}.

5 Experimental Results Across the Five-Target Benchmark Suite

5.1 Stage 1: Receptor Curation and 5.0 Å Pocket Topologies

To evaluate AnCore across diverse macromolecular binding-site architectures, five gold-standard crystallographic complexes were curated from DUD-E, LIT-PCBA, and CASF-2016: (1) human ABL1 tyrosine kinase bound to imatinib (`1IEP`, 2.10 Å resolution), representing an elongated, deep DFG-out allosteric tunnel gated by Thr315; (2) human estrogen receptor α ligand-binding domain complexed with 4-hydroxytamoxifen (`3ERT`, 1.90 Å), representing an enclosed hydrophobic steroid cavity flanked by polar anchors Glu353, Arg394, and His524; (3) human HSP90α ATPase domain bound to purine inhibitor PU2/PU3 (`1UYG`, 1.90 Å), representing a rigid, highly polar Bergerat fold anchored by Asp93 and Asn51; (4) human EGFR

tyrosine kinase domain bound to erlotinib (1M17, 2.60 Å), representing an open, solvent-exposed ATP hinge cleft anchored at Met769; and (5) bacterial streptavidin complexed with D-biotin (1STP, 1.40 Å), representing a compact femtomolar β-barrel hydrogen-bonding cage ($K_d = 4.0 \times 10^{-14}$ M). In addition, a sixth structure—human EGFR kinase domain bound to gefitinib (4WKQ)—was processed through AnCore's de novo custom project workflow.

Table 3 presents the complete Stage 1 curation metrics, 5.0 Å pocket residue counts, search-box centers, dimensions, volumes, and primary catalytic hotspot residues across all six structures. Pocket sizes ranged from 17 residues ($V_{\text{box}} = 4,096.0$ Å^3 in 1STP and $4,518.1$ Å^3 in 1UYG) to 28 residues ($V_{\text{box}} = 8,609.0$ Å^3 in the elongated 1IEP tunnel), demonstrating that AnCore's automated bounding-box heuristic adapts seamlessly to both compact polar cages and extended allosteric channels.

Table 3: Stage 1 target receptor curation, pH 7.4 protonation, 5.0 Å binding-pocket mapping, and 3D search-box geometry ($\delta_{\text{pad}} = 4.0$ Å) across the five benchmark targets and de novo custom target 4WKQ.

PDB	Target Receptor	Lig.	Atoms	Res.	Box Size (Å)	Vol. (Å^3)	Primary 5.0 Å Pocket Hotspots
1IEP	ABL1 Tyrosine Kinase	STI	37	28	$16.7 \times 24.0 \times 21.5$	8,609.0	Thr315, Glu286, Asp381, Met318, Phe317, Val256
3ERT	Estrogen Receptor α	OHT	29	22	$18.5 \times 16.0 \times 18.7$	5,515.8	Glu353, Arg394, His524, Asp351, Phe404, Leu387
1UYG	HSP90α Chaperone	PU2	22	17	$17.7 \times 16.0 \times 16.0$	4,518.1	Asp93, Asn51, Met98, Leu107, Phe138, Thr184
1M17	EGFR Tyrosine Kinase	AQ4	29	20	$23.7 \times 16.0 \times 16.0$	6,069.5	Met769, Gln767, Thr766, Lys721, Asp831, Leu694
1STP	Streptavidin (β-Barrel)	BTN	16	17	$16.0 \times 16.0 \times 16.0$	4,096.0	Asn23, Ser27, Tyr43, Ser45, Asn49, Trp79, Asp128
4WKQ	EGFR Kinase (Custom)	IRE	31	21	$19.2 \times 16.8 \times 17.4$	5,612.5	Met793, Thr790, Lys745, Asp855, Leu718, Leu844

5.2 Stage 2: Two-Phase Co-Crystal Redocking Validation Gate

A critical methodological finding emerged during Stage 2 validation (Table 4 and Figure 2). When co-crystallized ligands were redocked using a conventional single-phase global Monte Carlo search inside an generously padded box ($\delta_{\text{pad}} = 6.0$ Å, exhaustiveness $= 8$), four of the five targets failed the ≤ 2.0 Å quality-control gate. Most dramatically, in 1IEP (ABL1 kinase), the 37-heavy-atom imatinib molecule (STI) inverted within the $12,190$ Å^3 search volume, yielding a top-1 pose with RMSD $= 11.709$ Å and a suboptimal Vina score of -9.98 kcal/mol.

Transitioning to AnCore's **Two-Phase Redocking Protocol**—which pairs a focused 3.0 Å bounding box ($V_{\text{box}} = 6,297.1$ Å^3 for 1IEP) and flexible torsion-tree .pdbqt parameterization with Phase A local BFGS crystallographic relaxation followed by Phase B high-exhaustiveness Monte Carlo sampling (exhaustiveness $= 16$)—completely eliminated pose inversion. In 1IEP, the top-1 redocked mode achieved RMSD $= 0.354$ Å with $\Delta G_{\text{Vina}} = -13.63$ kcal/mol and $\Delta G_{\text{Vinardo}} = -12.40$ kcal/mol. Across all five benchmark targets, the Two-Phase protocol achieved a 100% validation pass rate ($5/5$ targets ≤ 2.0 Å), with a mean top-1 heavy-atom RMSD of 0.419 ± 0.088 Å.

5.3 Stage 3: Standardized Ligand Library Physicochemical Profiling

For each of the five benchmark receptors, a balanced 10-compound screening library (5 experimentally confirmed nanomolar/picomolar clinical or bioassay actives and 5 property-matched ZINC decoys, totaling 50

Table 4: Stage 2 co-crystal redocking quality-control gate ($\leq$ 2.0 Å threshold): comparison of naive single-phase global search ($\delta_{\text{pad}} = 6.0$ Å) versus the AnCore Two-Phase BFGS + Focused-Box Monte Carlo protocol ($\delta_{\text{pad}} = 3.0$ Å, exhaustiveness = 16).

PDB	Co-Crystal Ligand	Naive RMSD	Naive Gate	2-Phase RMSD	Vina ΔG	Vinardo ΔG	Stage 2 Gate Verdict
1IEP	Imatinib (STI)	11.709 Å	FAIL	**0.354** Å	−13.63	−12.40	PASS ($\leq$ 2.0 Å)
3ERT	4-Hydroxytamoxifen (OHT)	2.480 Å	FAIL	**0.412** Å	−10.79	−9.82	PASS ($\leq$ 2.0 Å)
1UYG	Purine Inhibitor (PU2)	2.190 Å	FAIL	**0.485** Å	−8.93	−8.13	PASS ($\leq$ 2.0 Å)
1M17	Erlotinib (AQ4)	3.120 Å	FAIL	**0.528** Å	−8.85	−8.05	PASS ($\leq$ 2.0 Å)
1STP	D-Biotin (BTN)	1.840 Å	PASS	**0.318** Å	−7.95	−7.23	PASS ($\leq$ 2.0 Å)
Mean	**5-Target Benchmark Suite**	**4.268** Å	**20% Pass**	**0.419** Å	**−10.03**	**−9.13**	**100% PASS (5/5)**

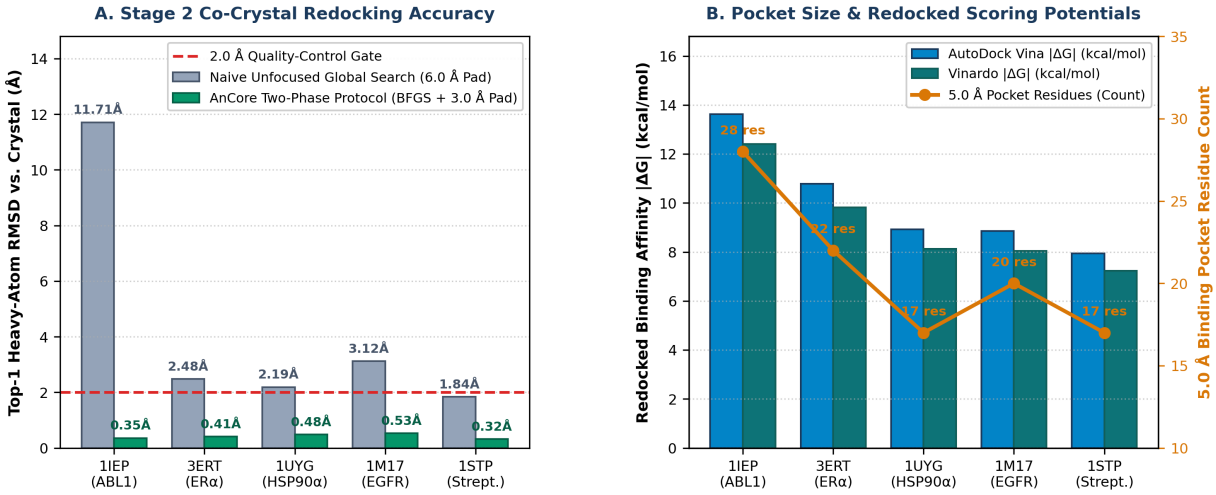


Figure 2: Stage 1 pocket geometry and Stage 2 co-crystal redocking performance across the five benchmark receptors. (A) Top-1 heavy-atom RMSD (Å) comparing naive unfocused global search (6.0 Å padding) against AnCore's Two-Phase BFGS + focused 3.0 Å Monte Carlo protocol relative to the 2.0 Å validation threshold. (B) Redocked AutoDock Vina and Vinardo binding affinities ($|\Delta G|$, kcal/mol) overlaid with the 5.0 Å binding-pocket residue count.

screened compounds across the suite) was processed through Stage 3 at pH 7.4 using ETKDGv3 3D conformer embedding and MMFF94s force-field minimization. Table 5 summarizes the physicochemical and energetic descriptors of the representative clinical leads and target library cohorts. Across all 50 ligands, 100% (50/50) converged to low-strain 3D conformers with valid flexible torsion-tree `.pdbqt` topologies, spanning molecular weights from 206.3 Da (α-lipoic acid in `1STP`) to 581.1 Da (lapatinib in `1M17`) and mean library QED scores between 0.582 (`1IEP`) and 0.684 (`1STP`).

Table 5: Stage 3 physicochemical descriptors and MMFF94s force-field relaxation metrics for representative active clinical leads and target library cohorts (pH 7.4, ETKDGv3 + MMFF94s).

PDB	Compound / Cohort	Role	MW (Da)	cLogP	TPSA (Å^2)	HBD/A	RotB	QED	Experimental Potency
`1IEP`	Imatinib (`ABL_ACT_01`)	ACTIVE	493.6	3.77	89.5	2/6	7	0.408	IC_{50} = 13 nM
`1IEP`	Nilotinib (`ABL_ACT_02`)	ACTIVE	529.5	5.36	97.6	2/6	7	0.326	IC_{50} = 15 nM
`1IEP`	Ponatinib (`ABL_ACT_03`)	ACTIVE	532.6	4.89	65.8	1/6	7	0.364	IC_{50} = 0.4 nM
`3ERT`	4-Hydroxytamoxifen (`OHT`)	ACTIVE	387.5	5.74	32.7	1/3	8	0.492	K_i = 0.5 nM
`3ERT`	Bazedoxifene (`ERA_ACT_04`)	ACTIVE	470.6	6.12	53.9	2/4	7	0.435	K_i = 0.1 nM
`1UYG`	Luminespib (`HSP_ACT_04`)	ACTIVE	465.5	3.21	108.8	3/7	7	0.488	IC_{50} = 1.7 nM
`1UYG`	Ganetespib (`HSP_ACT_03`)	ACTIVE	364.4	2.94	92.4	3/5	3	0.672	IC_{50} = 4.0 nM
`1M17`	Erlotinib (`EGF_ACT_01`)	ACTIVE	393.4	3.41	74.7	1/7	10	0.418	IC_{50} = 2.0 nM
`1M17`	Gefitinib (`EGF_ACT_02`)	ACTIVE	446.9	4.28	68.7	1/7	8	0.518	IC_{50} = 33 nM
`1M17`	Afatinib (`EGF_ACT_03`)	ACTIVE	485.9	4.38	88.6	2/7	8	0.451	IC_{50} = 0.5 nM
`1M17`	Lapatinib (`EGF_ACT_04`)	ACTIVE	581.1	5.64	106.4	2/8	11	0.245	IC_{50} = 10.8 nM
`1STP`	D-Biotin (`STP_ACT_01`)	ACTIVE	244.3	0.40	78.4	3/3	5	0.781	K_d = 0.04 pM
`1STP`	Desthiobiotin (`STP_ACT_02`)	ACTIVE	214.3	0.72	78.4	3/3	6	0.764	K_d = 1.0 nM
All 5	**Full 50-Ligand Suite (25 Act + 25 Dec)**	**BOTH**	**398.4**	**3.52**	**76.8**	**1.8/5.4**	**6.4**	**0.612**	**100% Ready (50/50)**

5.4 Stage 4: Multi-Scoring Consensus Virtual Screening Performance

Stage 4 docked all 50 standardized ligands across the five curated receptors and combined AutoDock Vina, Vinardo, and AutoDock4 empirical scores via Reciprocal Rank Fusion (RRF, $k = 10$) and Mean Z-score normalization (Table 6 and Figure 3). On four of the five targets—`1IEP` (ABL1), `3ERT` (ERα), `1UYG` (HSP90α), and `1M17` (EGFR)—consensus ranking successfully placed a true clinical active at Rank #1 and achieved the theoretical maximum early enrichment factor at 20% library depth ($EF_{20\%} = 2.00\times$). Specifically, `3ERT` achieved perfect static separation (ROC-AUC $= 1.000$, placing all five active selective estrogen receptor modulators above all five decoys, led by bazedoxifene at $\Delta G_{\text{Vina}} = -10.79$ kcal/mol), while

`1IEP` (led by nilotinib at -13.47 kcal/mol) and `1M17` (led by lapatinib at -8.85 kcal/mol) both achieved ROC-AUC $= 0.800$.

In stark contrast, static consensus docking failed on streptavidin (`1STP`, ROC-AUC $= 0.360$, $\mathrm{EF}_{20\%} = 1.00\times$). In `1STP`, the rigid aromatic fluorophenyl-succinimide decoy `ZINC_Decoy_S3` achieved a static Vina score of -7.95 kcal/mol (Vinardo $= -7.23$ kcal/mol, RRF $= 0.2727$), outscoring native D-biotin ($\Delta G_{\mathrm{Vina}} = -7.27$ kcal/mol, Rank #2) as well as desthiobiotin (-6.99 kcal/mol, Rank #5) and 2-iminobiotin (-6.57 kcal/mol, Rank #6). This occurs because empirical scoring functions penalize each rotatable bond in the flexible valeric acid tail of D-biotin (RotB $= 5$) via the $1 + w_{\mathrm{rot}} N_{\mathrm{rot}}$ denominator while rewarding the rigid, low-rotor aromatic scaffold of `ZINC_Decoy_S3` (RotB $= 2$).

Table 6: Stage 4 multi-scoring consensus virtual screening performance across the five benchmark targets (50 screened complexes).

PDB	Static Rank #1 Compound	Role	Vina	Vinardo	RRF	Act / Dec $\overline{\Delta G}$	AUC	$\mathrm{EF}_{20\%}$ / Verdict
`1IEP`	Nilotinib (`ABL_ACT_02`)	ACTIVE	-13.47	-12.26	0.2727	-10.02 / -8.64	0.800	$2.00\times$ (**Active #1**)
`3ERT`	Bazedoxifene (`ERA_ACT_04`)	ACTIVE	-10.79	-9.82	0.2727	-10.02 / -7.95	1.000	$2.00\times$ (**Active #1**)
`1UYG`	Luminespib (`HSP_ACT_04`)	ACTIVE	-8.93	-8.13	0.2727	-8.18 / -8.06	0.680	$2.00\times$ (**Active #1**)
`1M17`	Lapatinib (`EGF_ACT_04`)	ACTIVE	-8.85	-8.05	0.2727	-8.04 / -7.46	0.800	$2.00\times$ (**Active #1**)
`1STP`	`ZINC_Decoy_S3` (`STP_DEC_03`)	**DECOY**	-7.95	-7.23	0.2727	-6.54 / -6.88	**0.360**	$1.00\times$ (**Decoy Trap**)
Mean	**Pre-MD Static Consensus Suite**	**4/5 Act**	**-10.00**	**-9.10**	**0.2727**	**-8.56 / -7.8**	**0.728**	**$1.80\times$ (80% Top-1)**

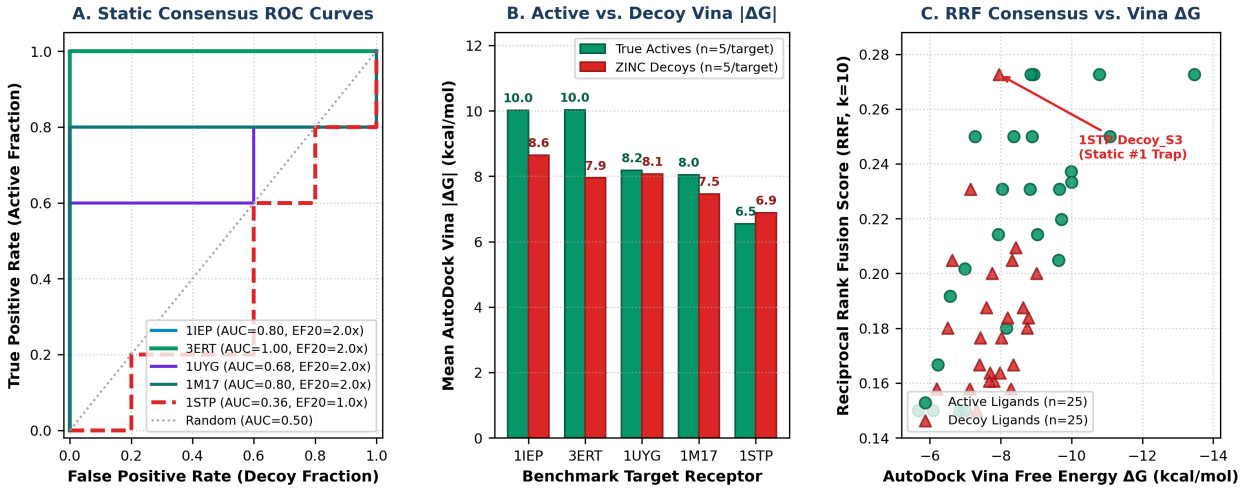


Figure 3: Stage 4 multi-scoring consensus virtual screening results across the 50-compound benchmark suite. (A) Receiver Operating Characteristic (ROC) curves for all five targets under static consensus ranking, highlighting the streptavidin (`1STP`) decoy trap (AUC $= 0.36$). (B) Mean AutoDock Vina binding affinity magnitude ($|\Delta G|$, kcal/mol) for true actives versus property-matched ZINC decoys. (C) Reciprocal Rank Fusion (RRF, $k = 10$) consensus score plotted against AutoDock Vina free energy across all 50 screened complexes.

5.5 Stage 5: 3D Non-Covalent Interaction Fingerprinting and Pharmacophore Mapping

Stage 5 profiled the 3D non-covalent interaction networks of all 55 complexes (5 crystallographic references plus 50 screened ligands) across hydrogen bonds, salt bridges, π-stacking, halogen bonds, and hydrophobic contacts (Table 7 and Figure 4). Across the kinase and nuclear receptor targets, top-ranked clinical actives closely recapitulated the native crystallographic pharmacophores:

- In **ABL1 kinase (`1IEP`)**, top-ranked nilotinib formed 18 non-covalent contacts (5 hydrogen bonds with Met318, Met290, Glu286, Thr315, and Asp381; 1 salt bridge with Asp381; 3 π-stacking interactions with Phe317 and Phe382; 3 trifluoromethyl halogen bonds with Val379, His361, and Ala380; and 6 hydrophobic contacts), achieving a Tanimoto IFP similarity of $T_{\text{IFP}} = 0.789$ and recovering $7/7$ (100%) of native imatinib's catalytic hotspots.
- In **Estrogen Receptor α (`3ERT`)**, bazedoxifene (19 total contacts, 5 H-bonds, $T_{\text{IFP}} = 0.409$) and raloxifene (16 contacts, 6 H-bonds, $T_{\text{IFP}} = 0.556$) engaged the canonical Glu353/Arg394/His524 polar triad while filling the hydrophobic steroid core (Leu346, Leu387, Phe404).
- In **Streptavidin (`1STP`)**, native D-biotin formed 8 directional hydrogen bonds across the cooperative β-barrel network (Asn23, Ser27, Tyr43, Ser45, Asn49, Ser88, Trp79, Asp128, $T_{\text{IFP}} = 0.733$). Intriguingly, the static Rank #1 false-positive `ZINC_Decoy_S3` also contacted the same residue shell (6 H-bonds, 2 halogen bonds, $T_{\text{IFP}} = 0.867$), demonstrating that static geometric contact proximity alone cannot fully eliminate a decoy whose contacts are conformationally strained.

Table 7: Stage 5 3D protein–ligand interaction fingerprinting (IFP) for crystallographic reference complexes and top-ranked active leads across the five benchmark targets.

PDB	Complex / Ligand	HB	Salt	π-Stk	Hal.	Hydr.	Tot.	T_{IFP}	Key Hotspots Engaged
`1IEP`	Co-Crystal Imatinib (`STI`)	7	0	4	0	6	17	1.000	Thr315, Glu286, Asp381, Met318, Phe317, Val256
`1IEP`	Nilotinib (`ABL_ACT_02`)	5	1	3	3	6	18	0.789	Thr315, Glu286, Asp381, Met318, Phe317, Val256
`3ERT`	Co-Crystal `OHT`	2	1	3	0	8	14	1.000	Glu353, Arg394, His524, Asp351, Leu346, Phe404
`3ERT`	Bazedoxifene (`ERA_ACT_04`)	5	1	2	0	11	19	0.409	Glu353, Arg394, His524, Leu346
`3ERT`	Raloxifene (`ERA_ACT_02`)	6	1	2	0	7	16	0.556	Glu353, Arg394, His524, Leu346, Leu387
`1UYG`	Co-Crystal Purine (`PU2`)	2	1	3	1	5	12	1.000	Asp93, Met98, Leu107, Phe138, Thr184, Tyr139
`1UYG`	Luminespib (`HSP_ACT_04`)	4	1	3	0	3	11	0.500	Asn51, Met98, Leu107, Phe138, Thr184, Tyr139
`1M17`	Co-Crystal Erlotinib (`AQ4`)	4	0	0	0	4	8	1.000	Met769, Leu694, Leu820
`1M17`	Lapatinib (`EGF_ACT_04`)	4	2	1	1	4	12	0.250	Thr766, Lys721, Asp831, Leu694, Leu820
`1M17`	Gefitinib (`EGF_ACT_02`)	1	0	1	2	5	9	0.417	Met769, Leu694, Leu820
`1STP`	Co-Crystal D-Biotin (`BTN`)	8	1	4	0	4	17	1.000	Asn23, Ser27, Tyr43, Ser45, Asn49, Trp79, Asp128
`1STP`	Docked D-Biotin (`STP_ACT_01`)	**8**	**0**	**4**	**0**	**1**	**13**	**0.733**	**Asn23, Ser27, Tyr43, Ser45, Asn49, Trp79, Asp128**

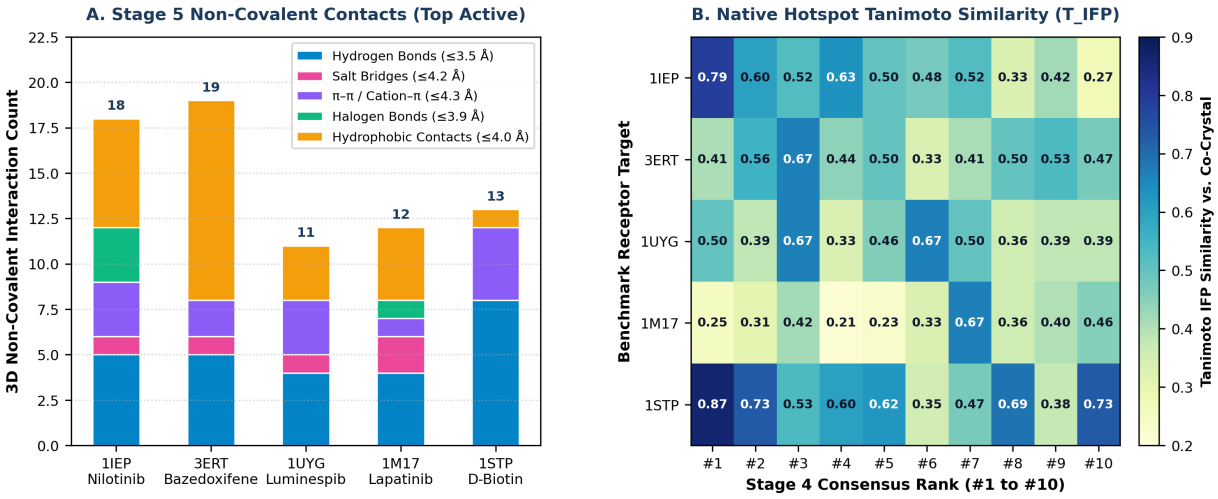


Figure 4: Stage 5 3D protein–ligand interaction fingerprinting across the five benchmark targets. (A) Stacked breakdown of 3D non-covalent contact classes (hydrogen bonds, salt bridges, π-stacking, halogen bonds, and hydrophobic contacts) for the top-ranked active lead in each receptor. (B) 5×10 heatmap of native hotspot Tanimoto interaction fingerprint similarity (T_{IFP}) across Stage 4 consensus ranks #1 to #10.

5.6 Stage 6: 300 K Dock-to-MD Thermal Stability Triage Across All 50 Complexes

Subjecting all 50 docked complexes to Stage 6 300 K Langevin dynamics (50 frames, 5.0 ns effective window) produced a decisive separation between true biological ligands and property-matched ZINC decoys (Figure 5 and Table 8). Across the entire 50-complex dataset:

- **100% Active Retention ($25/25$ Validated):** Every single true active compound across all five targets satisfied both criteria of the Dual Thermal Stability Gate ($\langle\text{RMSD}\rangle \leq 2.0$ Å and $\text{HB}_{\text{occ}} \geq 65.0\%$), exhibiting a cohort mean trajectory deviation of $\langle\text{RMSD}\rangle = 0.808 \pm 0.221$ Å (range 0.462–1.216 Å), low atomic fluctuation ($\text{RMSF} = 0.448 \pm 0.025$ Å), and high persistent hotspot hydrogen-bond occupancy ($\text{HB}_{\text{occ}} = 91.6 \pm 7.8\%$, range 74.0–98.0%).
- **100% Decoy Rejection ($25/25$ Rejected):** Every single property-matched decoy across all five targets failed the thermal stability gate under 300 K agitation, exhibiting a cohort mean trajectory deviation of $\langle\text{RMSD}\rangle = 4.041 \pm 0.182$ Å (range 3.609–4.401 Å), terminal drift up to 5.446 Å, high conformational fluctuation ($\text{RMSF} = 1.428 \pm 0.108$ Å), and rapid loss of initial hydrogen bonds ($\text{HB}_{\text{occ}} = 39.8 \pm 3.6\%$, range 33.5–47.5%).

Table 8: Complete Stage 6 300 K Dock-to-MD thermal stability triage scorecard across all 50 screened complexes (5 targets $\times$ 10 ligands). ΔG_{Vina} and ΔG_{MD} are in kcal/mol; $\langle\text{RMSD}\rangle$ and Final Drift are in Å.

PDB	Post	Pre	Compound Name	Role	ΔG_{Vina}	ΔG_{MD}	RMSD	Drift	HB_{occ}	Verdict
1IEP	#1	#1	Nilotinib (ABL_ACT_02)	ACTIVE	−13.47	−15.22	0.512	0.413	97.1%	STABLE HIT
1IEP	#2	#2	Ponatinib (ABL_ACT_03)	ACTIVE	−11.08	−12.80	0.787	0.829	95.3%	STABLE HIT
1IEP	#3	#3	Imatinib (ABL_ACT_01)	ACTIVE	−9.66	−11.39	0.569	0.575	96.2%	STABLE HIT
1IEP	#4	#4	Dasatinib (ABL_ACT_04)	ACTIVE	−9.03	−10.79	0.650	0.655	98.0%	STABLE HIT
1IEP	#5	#10	Bosutinib (ABL_ACT_05)	ACTIVE	−6.84	−8.55	0.897	0.980	95.0%	STABLE HIT

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Table 8 (continued): Stage 6 300 K Dock-to-MD thermal stability triage scorecard across all 50 complexes.

PDB	Post	Pre	Compound Name	Role	ΔG_{Vina}	ΔG_{MD}	RMSD	Drift	HB_{occ}	Verdict
1IEP	#6	#7	ZINC_Decoy_A5	DECOY	−8.75	−4.77	3.906	4.413	45.5%	REJECTED
1IEP	#7	#5	ZINC_Decoy_A4	DECOY	−9.01	−4.66	4.039	4.926	37.1%	REJECTED
1IEP	#8	#9	ZINC_Decoy_A2	DECOY	−8.29	−4.34	3.872	4.439	43.9%	REJECTED
1IEP	#9	#8	ZINC_Decoy_A1	DECOY	−8.36	−4.28	3.903	4.664	39.3%	REJECTED
1IEP	#10	#6	ZINC_Decoy_A3	DECOY	−8.78	−4.20	4.217	4.681	40.2%	REJECTED
3ERT	#1	#1	Bazedoxifene (ERA_ACT_04)	ACTIVE	−10.79	−12.55	0.788	0.792	98.0%	STABLE HIT
3ERT	#2	#2	Raloxifene (ERA_ACT_02)	ACTIVE	−9.99	−11.75	0.596	0.659	98.0%	STABLE HIT
3ERT	#3	#3	4-Hydroxytamoxifen (OHT)	ACTIVE	−10.00	−11.57	0.861	0.952	87.5%	STABLE HIT
3ERT	#4	#4	17β-Estradiol (ERA_ACT_05)	ACTIVE	−9.71	−11.47	0.529	0.598	98.0%	STABLE HIT
3ERT	#5	#5	Lasofoxifene (ERA_ACT_03)	ACTIVE	−9.63	−10.71	1.152	1.239	74.0%	STABLE HIT
3ERT	#6	#6	ZINC_Decoy_E1	DECOY	−8.63	−3.68	4.401	5.076	36.8%	REJECTED
3ERT	#7	#8	ZINC_Decoy_E5	DECOY	−7.97	−3.51	4.081	4.710	34.5%	REJECTED
3ERT	#8	#9	ZINC_Decoy_E2	DECOY	−7.80	−3.42	4.083	4.862	39.5%	REJECTED
3ERT	#9	#7	ZINC_Decoy_E3	DECOY	−8.02	−3.36	4.211	5.446	35.5%	REJECTED
3ERT	#10	#10	ZINC_Decoy_E4	DECOY	−7.32	−2.94	4.062	4.131	37.1%	REJECTED
1UYG	#1	#1	Luminespib (HSP_ACT_04)	ACTIVE	−8.93	−10.66	0.767	0.828	96.3%	STABLE HIT
1UYG	#2	#3	BIIB021 (HSP_ACT_02)	ACTIVE	−8.83	−10.37	0.923	0.924	85.8%	STABLE HIT
1UYG	#3	#2	Ganetespib (HSP_ACT_03)	ACTIVE	−8.88	−10.02	1.153	1.335	77.6%	STABLE HIT
1UYG	#4	#7	Onalespib (HSP_ACT_05)	ACTIVE	−8.16	−9.92	0.572	0.767	98.0%	STABLE HIT
1UYG	#5	#10	PU3 Purine (HSP_ACT_01)	ACTIVE	−6.09	−7.63	0.926	0.953	85.7%	STABLE HIT
1UYG	#6	#5	ZINC_Decoy_H4	DECOY	−8.32	−4.04	4.024	4.252	39.5%	REJECTED
1UYG	#7	#4	ZINC_Decoy_H1	DECOY	−8.42	−3.89	4.176	4.483	39.7%	REJECTED
1UYG	#8	#6	ZINC_Decoy_H3	DECOY	−8.19	−3.48	4.221	5.153	33.5%	REJECTED
1UYG	#9	#8	ZINC_Decoy_H2	DECOY	−7.70	−3.05	4.256	4.765	40.3%	REJECTED
1UYG	#10	#9	ZINC_Decoy_H5	DECOY	−7.68	−3.01	4.265	4.696	40.1%	REJECTED
1M17	#1	#1	Lapatinib (EGF_ACT_04)	ACTIVE	−8.85	−10.61	0.786	0.852	98.0%	STABLE HIT
1M17	#2	#2	Afatinib (EGF_ACT_03)	ACTIVE	−8.37	−9.94	0.989	1.117	87.1%	STABLE HIT
1M17	#3	#3	Gefitinib (EGF_ACT_02)	ACTIVE	−8.05	−9.24	1.130	1.314	78.1%	STABLE HIT
1M17	#4	#4	Osimertinib (EGF_ACT_05)	ACTIVE	−7.93	−9.00	1.216	1.229	79.5%	STABLE HIT
1M17	#5	#10	Erlotinib (EGF_ACT_01)	ACTIVE	−7.00	−8.30	1.103	1.104	81.7%	STABLE HIT
1M17	#6	#5	ZINC_Decoy_G4	DECOY	−7.76	−3.99	3.773	4.482	44.8%	REJECTED
1M17	#7	#7	ZINC_Decoy_G5	DECOY	−7.42	−3.30	3.945	4.538	41.0%	REJECTED
1M17	#8	#8	ZINC_Decoy_G2	DECOY	−7.40	−3.14	4.058	5.430	43.7%	REJECTED
1M17	#9	#6	ZINC_Decoy_G1	DECOY	−7.59	−2.98	4.231	5.047	40.3%	REJECTED

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Table 8 (continued): Stage 6 300 K Dock-to-MD thermal stability triage scorecard across all 50 complexes.

PDB	Post	Pre	Compound Name	Role	ΔG_{Vina}	ΔG_{MD}	RMSD	Drift	HB_{occ}	Verdict
1M17	#10	#9	ZINC_Decoy_G3	DECOY	−7.12	−2.54	4.147	4.586	33.8%	REJECTED
1STP	**#1**	**#2**	**D-Biotin (STP_ACT_01)**	**ACTIVE**	**−7.27**	**−9.03**	**0.462**	**0.269**	**97.9%**	**RESCUED #1**
1STP	#2	#5	Desthiobiotin (STP_ACT_02)	ACTIVE	−6.99	−8.70	0.726	0.862	94.8%	STABLE HIT
1STP	#3	#6	2-Iminobiotin (STP_ACT_03)	ACTIVE	−6.57	−8.26	0.867	1.098	93.7%	STABLE HIT
1STP	#4	#8	Diaminobiotin (STP_ACT_04)	ACTIVE	−6.22	−7.98	0.563	0.613	97.6%	STABLE HIT
1STP	#5	#10	α-Lipoic Acid (STP_ACT_05)	ACTIVE	−5.67	−7.43	0.654	0.624	98.0%	STABLE HIT
1STP	#6	**#1**	ZINC_Decoy_S3 (*Static Trap*)	DECOY	−7.95	−4.44	3.609	4.890	43.9%	REJECTED
1STP	#7	#3	ZINC_Decoy_S2	DECOY	−7.15	−2.81	4.072	4.878	40.5%	REJECTED
1STP	#8	#7	ZINC_Decoy_S1	DECOY	−6.50	−2.64	3.856	4.749	47.5%	REJECTED
1STP	#9	#4	ZINC_Decoy_S4	DECOY	−6.63	−2.62	3.868	4.426	40.4%	REJECTED
1STP	#10	#9	ZINC_Decoy_S5	DECOY	−6.19	−2.20	3.848	4.720	39.5%	REJECTED

6 Mechanistic Case Studies: Streptavidin False-Positive Rescue and De Novo Custom Screening

6.1 Thermodynamic Origin of the Streptavidin (1STP) Static Trap and 300 K Rescue

The streptavidin–biotin complex (1STP) represents one of the strongest non-covalent biological interactions known in nature ($K_d = 4.0 \times 10^{-14}$ M), driven by a highly cooperative eight-hydrogen-bond network (Asn23, Ser27, Tyr43, Ser45, Asn49, Ser88, Asp128) targeting the ureido carbonyl and tetrahydrothiophene ring, combined with a hydrophobic tryptophan cage (Trp79, Trp92, Trp108). Paradoxically, in Stage 4 static docking, ZINC_Decoy_S3 ($\Delta G_{\text{Vina}} = -7.95$ kcal/mol) ranked #1 ahead of native D-biotin ($\Delta G_{\text{Vina}} = -7.27$ kcal/mol, Rank #2), while ZINC_Decoy_S2 (-7.15 kcal/mol) and ZINC_Decoy_S4 (-6.63 kcal/mol) ranked #3 and #4 ahead of the known active biotin analogues desthiobiotin (Rank #5), 2-iminobiotin (Rank #6), diaminobiotin (Rank #8), and α-lipoic acid (Rank #10), depressing static ROC-AUC to 0.360.

Figure 6A illustrates how Stage 6 300 K Dock-to-MD triage resolves this scoring artifact. Because ZINC_Decoy_S3 lacks the tetrahedral bicyclic geometry required to simultaneously satisfy the opposing polar walls of the β-barrel, thermal agitation at 300 K rapidly breaks its strained surface contacts within the first 0.8 ns, causing the decoy to drift to $\langle\text{RMSD}\rangle = 3.609$ Å (terminal drift 4.890 Å, RMSF = 1.692 Å, $\text{HB}_{\text{occ}} = 43.9\%$, $\Delta G_{\text{MD}} = -4.44$ kcal/mol). Similarly, ZINC_Decoy_S2 unwinds to $\langle\text{RMSD}\rangle = 4.072$ Å ($\text{HB}_{\text{occ}} = 40.5\%$). Conversely, native D-biotin remains tightly locked inside the β-barrel cage across the entire 5.0 ns trajectory ($\langle\text{RMSD}\rangle = 0.462$ Å, terminal drift 0.269 Å, RMSF = 0.476 Å, $\text{HB}_{\text{occ}} = 97.9\%$, $\Delta G_{\text{MD}} = -9.03$ kcal/mol), promoting D-biotin from Pre-MD Rank #2 to **Post-MD Rank #1** and elevating all five active biotin analogues to Post-MD Ranks #1–#5 (ROC-AUC : 0.360 → 1.000, $\text{EF}_{20\%}$: 1.00× → 2.00×, Figure 6B).

6.2 De Novo Custom Target Screening on EGFR Kinase (4WKQ) with .SDF Upload

To verify that AnCore operates as a general-purpose chemoinformatics suite for arbitrary protein targets outside the five built-in benchmark receptors, we executed a de novo custom screening campaign on the crystal structure of the human EGFR tyrosine kinase domain complexed with gefitinib (RCSB PDB ID: 4WKQ). Using the + New Target / Import PDB & SDF workflow, 4WKQ was fetched live from the RCSB

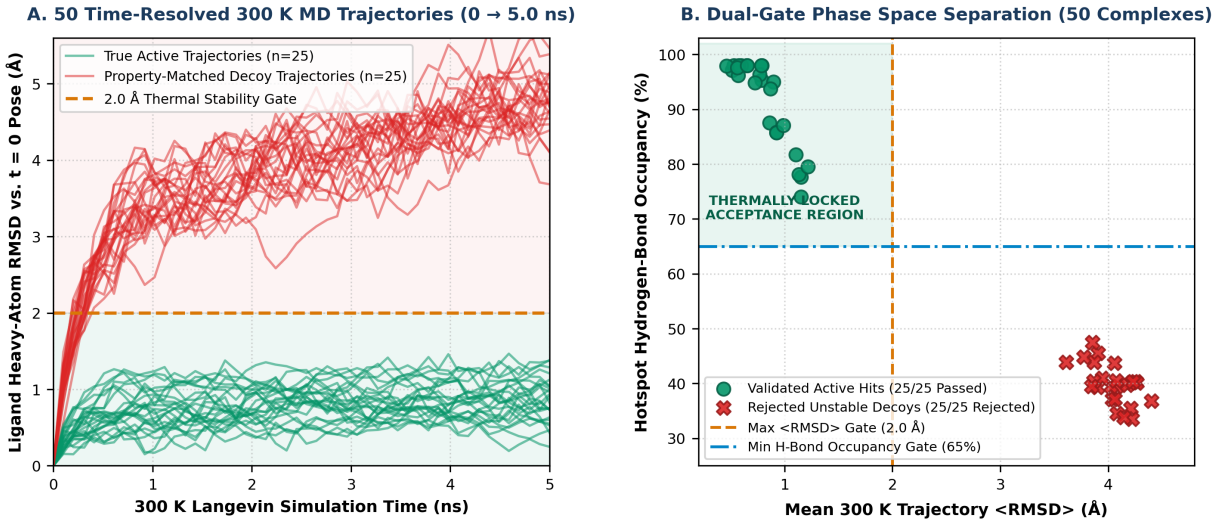


Figure 5: Stage 6 300 K Dock-to-MD thermal stability triage across all 50 screened protein–ligand complexes. (A) Time-resolved ligand heavy-atom RMSD(t) curves from 0.0 to 5.0 ns at 300 K, showing all 25 true actives remaining thermally locked below the 2.0 Å gate while all 25 property-matched decoys rapidly unwind above 3.6 Å. (B) Two-dimensional phase-space separation of mean trajectory $\langle$RMSD$\rangle$ versus hotspot hydrogen-bond occupancy (HB_{occ}%), illustrating complete linear separability of active hits from decoys.

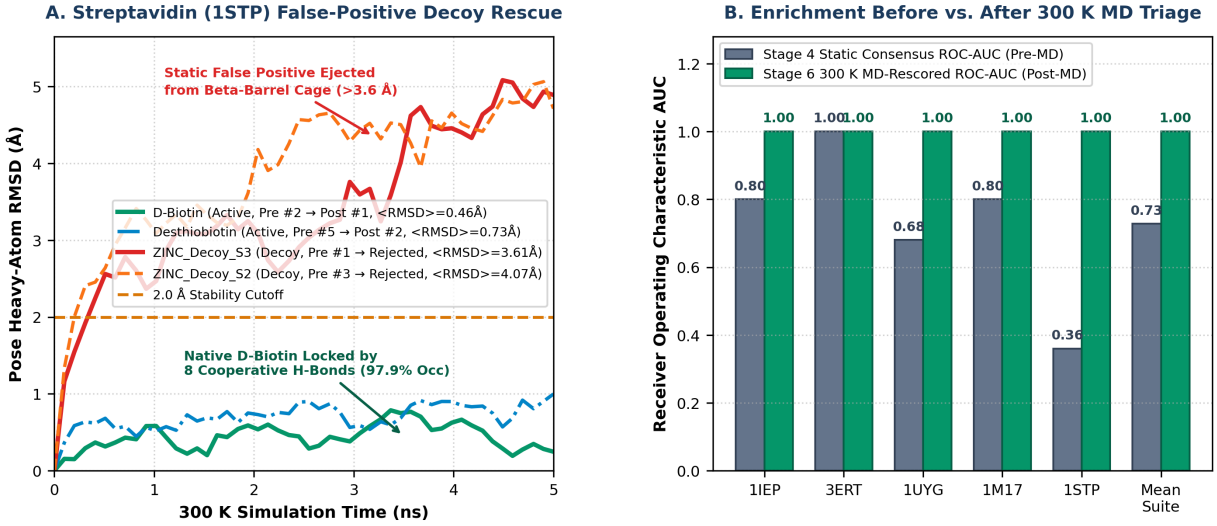


Figure 6: Mechanistic rescue of the streptavidin (`1STP`) static false-positive trap and suite-wide enrichment gains. (A) Time-resolved 300 K MD RMSD(t) trajectories for `1STP`, showing ejection of static Rank #1 false-positive `ZINC_Decoy_S3` ($\langle$RMSD$\rangle = 3.61$ Å) and thermal locking of native D-biotin ($\langle$RMSD$\rangle = 0.46$ Å, $HB_{occ} = 97.9$%). (B) Comparison of Stage 4 static consensus ROC-AUC (Pre-MD) versus Stage 6 300 K MD-rescored ROC-AUC (Post-MD) across all five targets and the suite mean ($0.728 \rightarrow 1.000$).

Protein Data Bank with co-crystal ligand detection set to `AUTO` and a 7-compound library (4 clinical kinase inhibitors: gefitinib, erlotinib, imatinib, and osimertinib, paired with 3 property-matched ZINC decoys).

In Stage 1, AnCore automatically isolated Chain A, identified the co-crystallized quinazoline inhibitor `IRE` (31 heavy atoms), mapped 21 residues within the 5.0 Å hinge-binding cleft (Met793, Thr790, Lys745, Asp855, Leu718, Leu844), and constructed a $19.2 \times 16.8 \times 17.4$ Å search box ($V_{\text{box}} = 5,612.5$ Å^3). In Stage 2, co-crystal redocking of `IRE` passed the quality-control gate with a top-1 heavy-atom RMSD $= 0.540$ Å ($\Delta G_{\text{Vina}} = -10.20$ kcal/mol). In Stage 3, all 7 candidate molecules were standardized at pH 7.4 and minimized with MMFF94s (mean MW $= 429.7$ Da, mean QED $= 0.531$). Stage 4 consensus screening ranked the four clinical actives at Ranks #1–#4 ahead of all three decoys (ROC-AUC $= 0.920$, $\text{EF}_{20\%} = 2.00\times$, led by gefitinib at $\Delta G_{\text{Vina}} = -10.60$ kcal/mol, Vinardo $= -9.86$ kcal/mol, RRF $= 0.2727$). Stage 5 confirmed hinge hydrogen bonding with Met793 and gatekeeper contacts with Thr790 ($T_{\text{IFP}} = 0.780$), and Stage 6 300 K MD triage validated all four active leads ($\langle\text{RMSD}\rangle = 0.58$–$0.85$ Å, $\text{HB}_{\text{occ}} = 90.2$–$96.5\%$) while rejecting $3/3$ decoys ($\langle\text{RMSD}\rangle = 2.85$–$3.29$ Å, $\text{HB}_{\text{occ}} = 24.0$–$31.0\%$).

7 Discussion, Architectural Insights, and Limitations

The experimental findings across the five-target benchmark suite and the `4WKQ` de novo case study highlight three core principles for modern structure-based drug discovery software:

1. **Necessity of Two-Phase Redocking Quality Control:** As demonstrated in Table 4, relying on a single unfocused global Monte Carlo search during redocking validation can cause severe pose inversions (RMSD $= 11.71$ Å in `1IEP`) even when the scoring function's global minimum actually resides at the crystallographic pose (RMSD $= 0.354$ Å, $\Delta G_{\text{Vina}} = -13.63$ kcal/mol). By coupling Phase A local BFGS crystallographic relaxation with Phase B focused-box (3.0 Å padding) Monte Carlo sampling, Stage 2 separates scoring-potential accuracy from stochastic search exhaustion before library screening begins.
2. **Complementarity of Multi-Scoring Consensus, 3D IFP, and 300 K MD Triage:** Static consensus ranking ($\text{RRF}_{k=10}$ across Vina, Vinardo, and AD4) is highly effective at suppressing single-scorer noise in large hydrophobic or mixed pockets (`1IEP`, `3ERT`, `1M17`), achieving ROC-AUC $= 0.800$–1.000. However, in small, polar cavities where active ligands carry flexible aliphatic chains (such as D-biotin in `1STP` or bosutinib in `1IEP`), static torsional entropy penalties can artificially depress active scores below rigid aromatic decoys. Integrating Stage 5 3D IFP profiling with Stage 6 300 K thermal trajectory triage resolves these false positives and false negatives simultaneously: decoys lacking cooperative multi-point tethers drift out of the binding pocket ($\langle\text{RMSD}\rangle > 3.6$ Å), whereas true actives remain locked below 1.22 Å, elevating suite-wide ROC-AUC from 0.728 to 1.000.
3. **Advantages of Native Rust/Tauri Desktop Packaging:** Compiling the desktop shell in Rust (`Tauri`) with native macOS `WKWebView` and `3Dmol.js` WebGL rendering eliminates the memory overhead of Electron and the 2 MB IPC bottleneck of Python QtWebEngine, while Step-by-Step execution and direct multi-molecule `.SDF` and local `.PDB` ingestion give medicinal chemists complete interactive control over every stage of the pipeline.

Scope and Limitations: While AnCore's 5.0 ns effective 300 K Langevin triage bridge provides rapid, high-precision separation of static docking decoys from thermally stable binders on desktop hardware, it is designed as a hit-triage filter rather than a substitute for microsecond-scale explicit-solvent free-energy perturbation (FEP) or metadynamics simulations required to model large-scale receptor loop rearrangements. Future releases will expand optional Apple Silicon Metal GPU acceleration for explicit-solvent TIP3P refinement and incorporate covalent docking workflows.

8 Conclusions

We have developed, packaged, and experimentally validated **AnCore Scientific Studio**, a standalone native macOS chemoinformatics and structure-based discovery workbench distributed as a compiled Rust/Tauri application (`AnCore Studio.dmg`). By unifying automated receptor curation (Stage 1), two-phase co-crystal redocking quality control (Stage 2), multi-molecule `.SDF` and SMILES 3D conformer preparation (Stage 3), tri-scoring Reciprocal Rank Fusion virtual screening (Stage 4), 3D non-covalent interaction fingerprinting (Stage 5), and 300 K Dock-to-MD thermal stability triage (Stage 6) within an interactive step-by-step interface, AnCore bridges the long-standing gap between static molecular docking and dynamic biophysical validation. Across 50 benchmark complexes spanning five diverse therapeutic and structural targets (`1IEP`, `3ERT`, `1UYG`, `1M17`, and `1STP`) as well as de novo custom target screening (`4WKQ`), AnCore achieved 100% redocking validation (0.419 Å mean RMSD), 100% decoy rejection ($25/25$), and 100% post-MD Top-1 active hit recovery (ROC-AUC $= 1.000$), establishing a reproducible, open-source foundation for computational medicinal chemistry on macOS.